

MILESTONE 1

Platform Foundation & Authentication System

Enterprise Learning Platform – Skill & Career Guidance System

1. Milestone Overview

Milestone 1 establishes the UI, database, and secure authentication foundation of the Enterprise Learning Platform.

Key Features

- Landing/Home Page
- Student Registration & Login
- Form Validation
- BCrypt Password Hashing
- MySQL User Management
- JWT Authentication
- JWT Authentication Middleware
- Spring Security
- Role-Based Authorization
- Protected REST APIs
- Logout & Token Management
- Google Authentication Interface
- Responsive UI/UX

Architecture: React.js → REST API → Spring Boot → Spring Security → JPA/Hibernate → MySQL

2. Project Objective

To provide a secure authentication foundation for future platform modules such as:

- Career Guidance
- Skill Assessments
- Learning Roadmaps
- Courses
- Internships
- Badges & Certificates

- Career Recommendations
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3. Landing Page & Authentication

Landing Page

- Home, Features, Roadmaps, About
- Login & Get Started
- Course Modules
- Career Roadmaps
- QR Certificates
- Module Badges

Registration

- Full Name
- Email
- Password
- Confirm Password
- Terms & Conditions
- Google Authentication

Login

- Email & Password
- Student Role
- Show/Hide Password
- Remember Me
- Forgot Password
- Google Authentication

Validation

- Required fields
 - Email format
 - Password strength
 - Password matching
 - Invalid credential handling
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4. Database & Password Security

User data is stored in MySQL.

Field	Description
ID	User ID
Full Name	Student name
Email	Email address
Password	BCrypt hash
Role	Student/Admin/Mentor
Status	Account status
Created/Updated	Timestamps

Security: Passwords are never stored in plain text. BCrypt hashing is used before database storage.

5. JWT Authentication

JWT is used to authenticate users and protect private APIs.

Authentication Flow

Login

↓

Spring Security

↓

MySQL User Verification



BCrypt Password Verification



JWT Generation



JWT Returned to Client



Protected API Request



JWT Verification



Access Granted

JWT Contains

- **sub** – User ID
- **role** – User role
- **iat** – Issued time
- **exp** – Expiration time

Passwords and JWT signing secrets are never exposed in the token/frontend.

6. JWT Authentication Middleware

The JWT middleware/filter:

1. Extracts JWT from the request.
2. Validates the token.
3. Checks expiration.
4. Extracts user identity.
5. Verifies the user role.
6. Allows or rejects the request.

Request



JWT Filter



Token Validation



User + Role Verification



Protected Controller



Service → Repository → MySQL

7. Spring Security & Authorization

Spring Security protects backend APIs.

Current Role

- STUDENT

Future Roles

- ADMIN
- MENTOR

JWT → Authentication → Role Check → API Access



Unauthorized

Requests with a missing, invalid, expired, or unauthorized JWT are rejected.

8. Protected APIs

JWT authentication protects:

- Student Profile
- Dashboard
- Skill Assessments
- Course Modules
- Learning Progress
- Career Roadmaps
- Internships
- Badges
- Certificates

Authentication APIs

Method	Endpoint
POST	<code>/api/auth/register</code>
POST	<code>/api/auth/login</code>

9. Logout & Google Authentication

Logout

- Clear authentication state
- Remove token
- Redirect to Login/Home
- JWT expiration prevents indefinite access

Google Authentication

Google Login



Google Verification



Backend



Create/Retrieve User



Authenticated Session

The architecture is prepared for Google OAuth integration.

10. UI/UX & Responsive Design

The platform provides:

- Modern and clean UI
- Green/mint theme
- Rounded cards and inputs
- Consistent typography
- Responsive layouts
- Accessible controls

Supports:

Desktop | Laptop | Tablet | Mobile

11. Final Outcome

Milestone 1 delivers a secure authentication foundation for the Enterprise Learning Platform:

React UI → Spring Boot → Spring Security → JWT → Role Authorization → Protected APIs → MySQL

This foundation enables future development of courses, assessments, roadmaps, internships, badges, certificates, and career recommendations.